

VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Practical No. 06

Aim: Implementation of Hierarchical Clustering.

Software Required: Python IDE, Anaconda, VS Code, Jupyter Nootbook, Google Colab.

Theory:

Hierarchical clustering

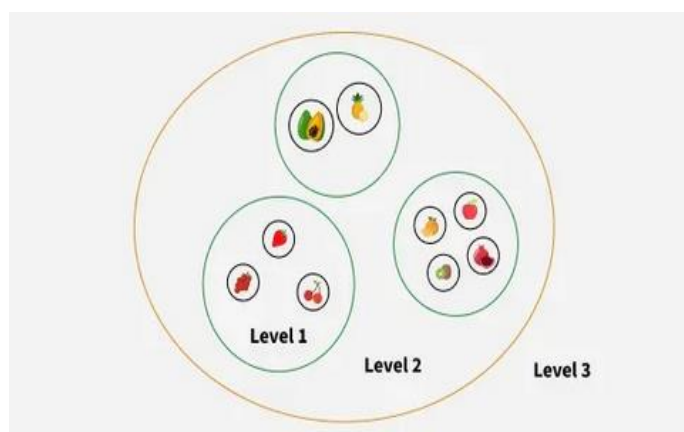
Hierarchical clustering is used to group similar data points together based on their similarity creating a hierarchy or tree-like structure. The key idea is to begin with each data point as its own separate cluster and then progressively merge or split them based on their similarity.

Let's understand this with the help of an example

Imagine you have four fruits with different weights: an apple (100g), a banana (120g), a cherry (50g) and a grape (30g). Hierarchical clustering starts by treating each fruit as its own group.

- It then merges the closest groups based on their weights.
- First the cherry and grape are grouped together because they are the lightest.
- Next the apple and banana are grouped together.

Finally, all the fruits are merged into one large group, showing how hierarchical clustering progressively combines the most similar data points.



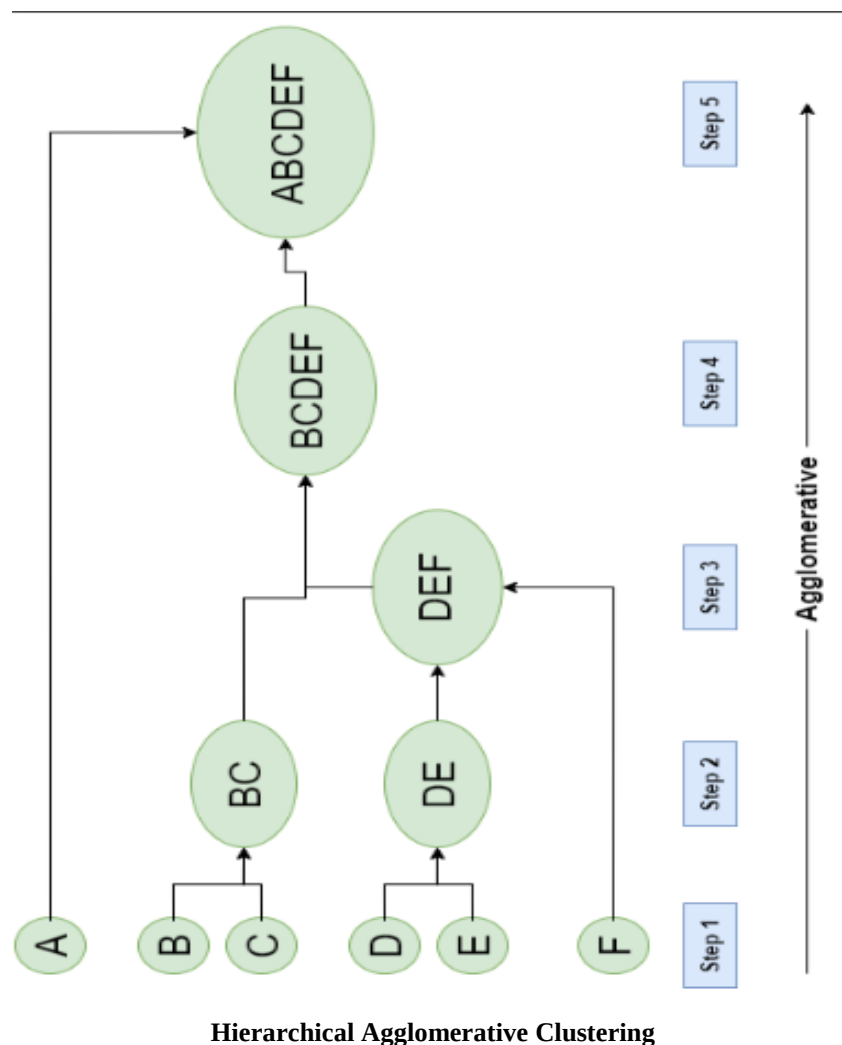
Types of Hierarchical Clustering

Now we understand the basics of hierarchical clustering. There are two main types of hierarchical clustering.

1. Agglomerative Clustering
2. Divisive clustering

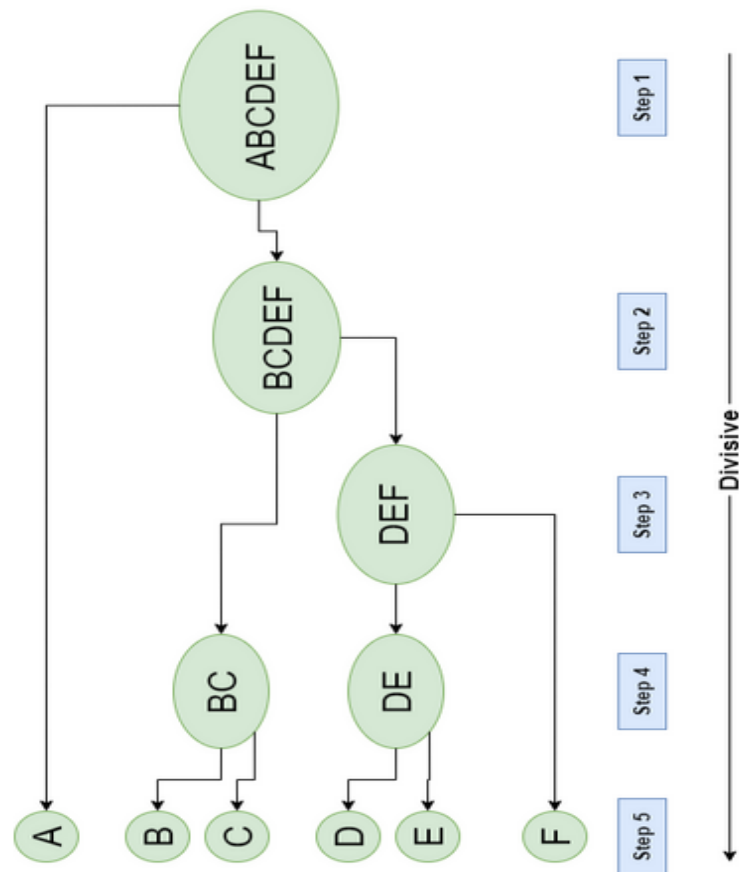
Hierarchical Agglomerative Clustering

It is also known as the bottom-up approach or hierarchical agglomerative clustering (HAC). Unlike flat clustering hierarchical clustering provides a structured way to group data. This clustering algorithm does not require us to prespecify the number of clusters. Bottom-up algorithms treat each data as a singleton cluster at the outset and then successively agglomerate pairs of clusters until all clusters have been merged into a single cluster that contains all data.



Hierarchical Divisive clustering

It is also known as a **top-down approach**. This algorithm also does not require to prespecify the number of clusters. Top-down clustering requires a method for splitting a cluster that contains the whole data and proceeds by splitting clusters recursively until individual data have been split into singleton clusters.

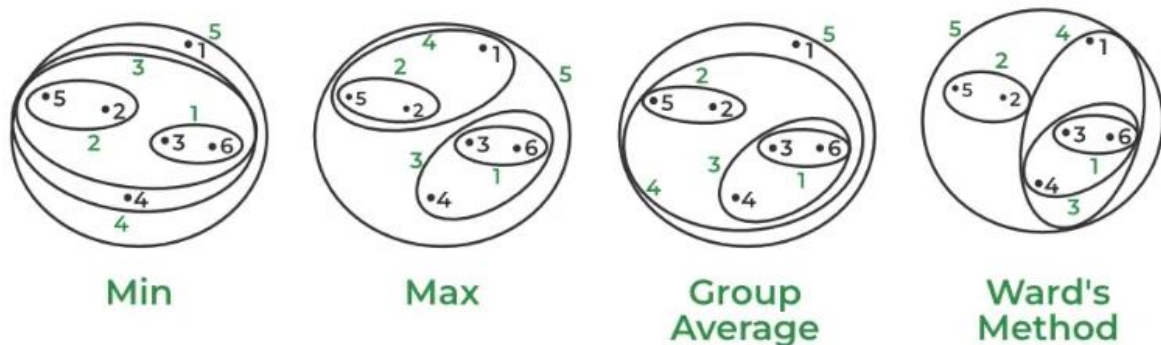


Computing Distance Matrix

While merging two clusters we check the distance between two every pair of clusters and merge the pair with the least distance/most similarity. But the question is how is that distance determined. There are different ways of defining Inter Cluster distance/similarity. Some of them are:

1. **Min Distance:** Find the minimum distance between any two points of the cluster.
2. **Max Distance:** Find the maximum distance between any two points of the cluster.
3. **Group Average:** Find the average distance between every two points of the clusters.
4. **Ward's Method:** The similarity of two clusters is based on the increase in squared error when two clusters are merged.

Distance Matrix Comparision in Hierarchical Clustering



Distance Matrix Comparision in Hierarchical Clustering

Applications of Agglomerative Clustering

You can find application of Agglomerative Clustering in many areas of unsupervised machine learning tasks. The following are some important areas of its applications in machine learning –

1) Image Segmentation

- Agglomerative clustering is used in computer vision to group pixels of an image based on color, intensity, or texture.
- Example: Dividing a medical image (like an MRI scan) into regions such as tissues, bones, and tumors.

2) Document Clustering

- It can group similar documents or articles together by analyzing the words they contain.
- Example: News articles can be clustered into topics like politics, sports, technology, etc.

3) Customer Behaviour Analysis (Customer Segmentation)

- Businesses can group customers with similar buying behavior to target them better.
- Example: Online shopping sites cluster users based on purchase history—like “electronics lovers,” “fashion buyers,” or “frequent shoppers.”

4) Market Segmentation

- Companies use it to divide the entire market into groups of people with similar needs or characteristics.
- Example: In the mobile phone market, customers may be segmented into “budget phone users,” “camera lovers,” and “high-performance seekers.”

5) Social Network Analysis

- It helps in finding communities or groups of people within social networks.
- Example: On Facebook, clustering can group users who interact frequently (friends, classmates, or colleagues).

Advantages of Agglomerative Clustering

Following are the advantages of using Agglomerative Clustering –

- Produces a dendrogram that shows the hierarchical relationship between the clusters.
- Can handle different types of distance metrics and linkage methods.
- Allows for a flexible number of clusters to be extracted from the data.
- Can handle large datasets with efficient implementations.

Disadvantages of Agglomerative Clustering

Following are some of the disadvantages of using Agglomerative Clustering –

- Can be computationally expensive for large datasets.
- Can produce imbalanced clusters if the distance metric or linkage method is not appropriate for the data.
- The final result may be sensitive to the choice of distance metric and linkage method used.
- The dendrogram may be difficult to interpret for large datasets with many clusters.

Program:

1. Python implementation of the above algorithm using the scikit-learn library:

```
from sklearn.cluster import AgglomerativeClustering
import numpy as np
X = np.array([[1, 2], [1, 4], [1, 0],
              [4, 2], [4, 4], [4, 0]])
clustering = AgglomerativeClustering(n_clusters=2).fit(X)
print(clustering.labels_)
```

Output:

```
[1, 1, 1, 0, 0, 0]
```

2. Implementation code for Distance Matrix Comparision

```
import numpy as np
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

X = np.array([[1, 2], [1, 4], [1, 0],
```

```
[4, 2], [4, 4], [4, 0]])
```

```
Z = linkage(X, 'ward') # Ward Distance
```

```
dendrogram(Z) #plotting the dendrogram
```

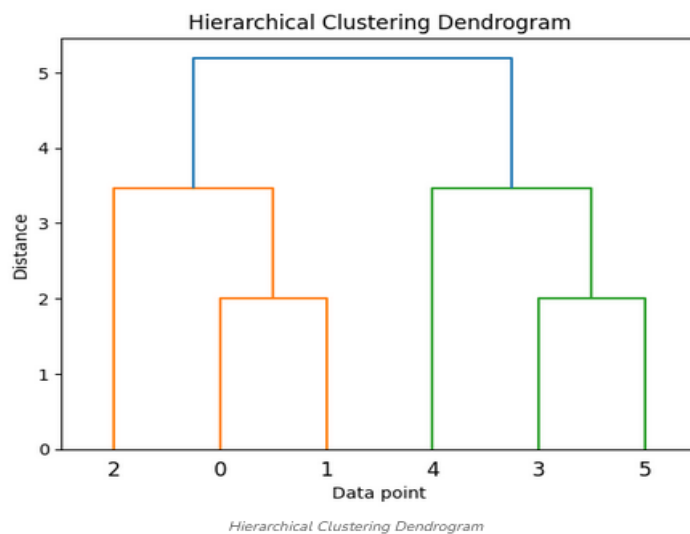
```
plt.title('Hierarchical Clustering Dendrogram')
```

```
plt.xlabel('Data point')
```

```
plt.ylabel('Distance')
```

```
plt.show()
```

Output:



Conclusion: Hierarchical clustering helps to understand how data can be grouped step by step, showing clear cluster relationships through a dendrogram and making data analysis easier to interpret.